

WhatWeEatInAmerica

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1 Stratifying “What we eat in America”

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We apply Dleto stratification on tensors emerging from nutrition data. We use a USDA Government survey

[What We Eat in America](#), NHANES 2017-March 2020 Prepandemic

The data consists of several thousand foods sold to the public within the USA with each item assigned measures of multiple food types. For example, what fraction of a serving consists of real fruits, vegetables, grains, added sugars and etc.

DISCLAIMER. This notebook is intended to illustrate Dleto stratification algorithms only and does not represent any suggested or implied nutritional advice nor does it represent a complete understanding of the underlying significance of the data.

-
- Loading the Data
 - Creating Food Categories
 - A What We Eat Tensor
 - Stratifying the WWE Tensor
 - Analysis

1.1 1. Loading the data

We begin by loading the data and preparing for analysis.

Let us start by loading the package we will use. In some situations you may need to add the package to your Julia system by uncommenting the relevant commands. This is typically a one-time step and can be avoided in future runs.

```
[123]: using Pkg
      # Uncomment to install packages if needed.
      Pkg.add("CSV"); Pkg.add("DataFrames")
      using CSV, DataFrames
      Pkg.activate("../") # Activate the main project
      using Dleto
      using Plots
```

```

Resolving package versions...
Compat entries added for
Project No packages added to or removed from
`~/CODE/OpenDleto/Project.toml`
Manifest No packages added to or removed from
`~/CODE/OpenDleto/Manifest.toml`
Precompiling packages...
2469.6 ms    Dleto
1 dependency successfully precompiled in 5 seconds. 324 already precompiled.
1 dependency precompiled but a different version is currently

```

loaded. Restart julia to access the new version. Otherwise, loading dependents of this package may trigger further precompilation to work with the unexpected version.

```

Resolving package versions...
Compat entries added for
Project No packages added to or removed from
`~/CODE/OpenDleto/Project.toml`
Manifest No packages added to or removed from
`~/CODE/OpenDleto/Manifest.toml`
Activating project at `~/CODE/OpenDleto`

```

Note. The USDA data set is provided as a single Microsoft Excel formatted spreadsheet consisting of two sheets: the data and a key. For convenience the file is stored locally as two separate CSV files, one for each sheet.

```

[124]: # Load the data and key files
data = CSV.read("FPED_1720.csv", DataFrame)
key_data = CSV.read("FPED_1720-key.csv", DataFrame)

println("Total number of foods: $(size(data,1))")
println("Number of Categories: $(size(key_data,1))")
println("\nSample of food items:")
println(data[147:155, vcat(1:2, 33:34, 38)]) # Show first few items and columns

```

Total number of foods: 7444

Number of Categories: 39

Sample of food items:

9×5 DataFrame

Row	FOODCODE	DESCRIPTION
D_MILK (cup eq)	D_YOGURT (cup eq)	ADD_SUGARS (tsp eq)
Int64	String	
Float64	Float64	Float64

1	11513802	Chocolate milk, made from light ...	0.36
0.0		0.85	
2	11513803	Chocolate milk, made from light ...	0.36
0.0		0.85	
3	11513804	Chocolate milk, made from light ...	0.36
0.0		0.85	
4	11513805	Chocolate milk, made from light ...	0.18
0.0		1.61	
5	11513850	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
6	11513851	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
7	11513852	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
8	11513853	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
9	11513854	Chocolate milk, made from sugar ...	0.36
0.0		0.0	

Number of Categories: 39

Sample of food items:

9×5 DataFrame

Row	FOODCODE	DESCRIPTION
D_MILK (cup eq)	D_YOGURT (cup eq)	ADD_SUGARS (tsp eq)
Int64	String	
Float64	Float64	Float64

1	11513802	Chocolate milk, made from light ...	0.36
0.0		0.85	
2	11513803	Chocolate milk, made from light ...	0.36
0.0		0.85	
3	11513804	Chocolate milk, made from light ...	0.36
0.0		0.85	
4	11513805	Chocolate milk, made from light ...	0.18
0.0		1.61	
5	11513850	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
6	11513851	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
7	11513852	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
8	11513853	Chocolate milk, made from sugar ...	0.36
0.0		0.0	
9	11513854	Chocolate milk, made from sugar ...	0.36

0.0 0.0

More detail about the labels may be obtained by inspecting them individually. For instance, a food product that includes “light syrup” has sugar as an added ingredient, whereas “sugar free syrup” adds 0.0 sugar.

```
[125]: println( data[147,2])
println( data[155,2])
```

Chocolate milk, made from light syrup with reduced fat milk
Chocolate milk, made from sugar free syrup with fat free milk

To understand the 39 categories we can access the `key_data`.

```
[126]: # Examine the key data to understand the nutritional categories
println("Key data structure:")
println(key_data)
```

Key data structure:

39×4 DataFrame

Row	column	variable_name		
		variable_description	units	
		String3	String31	String
		String15?		
1	A	FOODCODE	Food code	
missing				
2	B	DESCRIPTION	Food description	
missing				
3	C	F_TOTAL	Total intact or cut fruits and f...	cup eq
4	D	F_CITMLB	Intact fruits (whole or cut) of ...	cup eq
5	E	F_OTHER	Intact fruits (whole or cut); ex...	cup eq
6	F	F_JUICE	Fruit juices, citrus and non cit...	cup eq
7	G	V_TOTAL	Total dark green, red and orange...	cup eq
8	H	V_DRKGR	Dark green vegetables	cup eq
9	I	V_REDOR_TOTAL	Total red and orange vegetables ...	cup eq
10	J	V_REDOR_TOMATO	Tomatoes and tomato products	cup eq
11	K	V_REDOR_OTHER	Other red and orange vegetables,...	cup eq
12	L	V_STARCHY_TOTAL	Total starchy vegetables (white ...	cup eq
13	M	V_STARCHY_POTATO	White potatoes	cup eq
14	N	V_STARCHY_OTHER	Other starchy vegetables, exclud...	cup eq
15	O	V_OTHER	Other vegetables not in the vege...	cup eq
16	P	V_LEGUMES	Legumes computed as vegetables	cup eq
17	Q	G_TOTAL	Total whole and refined grains	oz eq
18	R	G_WHOLE	Whole grains	oz eq
19	S	G_REFINED	Refined or non-whole grains	oz eq

20	T	PF_TOTAL	Total meat, poultry, seafood, or...	oz eq
21	U	PF_MPS_TOTAL	Total meat, poultry, seafood, or...	oz eq
22	V	PF_MEAT	Beef, veal, pork, lamb, game mea...	oz eq
23	W	PF_CUREDMEAT	Cured/luncheon meat made from be...	oz eq
24	X	PF_ORGAN	Organ meat from beef, veal, pork...	oz eq
25	Y	PF_POULT	Chicken, turkey, Cornish hens, a...	oz eq
26	Z	PF_SEAFD_HI	Seafood (finfish, shellfish and ...	oz eq
27	AA	PF_SEAFD_LOW	Seafood (finfish, shellfish and ...	oz eq
28	AB	PF_EGGS	Eggs (chicken, duck, goose, quai...	oz eq
29	AC	PF_SOY	Soy products, excluding calcium ...	oz eq
30	AD	PF_NUTSDS	Peanuts, tree nuts, and seeds, e...	oz eq
31	AE	PF_LEGUMES	Legumes computed as protein foods	oz eq
32	AF	D_TOTAL	Total milk, yogurt, cheese, and ...	cup eq
33	AG	D_MILK	Fluid milk and calcium fortified...	cup eq
34	AH	D_YOGURT	Yogurt	cup eq
35	AI	D_CHEESE	Cheese	cup eq
36	AJ	OILS	Oils	grams
37	AK	SOLID_FATS	Solid fats	grams
38	AL	ADD_SUGARS	Foods defined as added sugars	tsp eq
39	AM	A_DRINKS	Alcoholic beverages	no. of

drinks

To prepare for further analysis, we do an initial analysis that identifies missing values and total range of values.

```
[127]: using Statistics

# Extract the numerical data for tensor creation
# Remove the first two columns (FOODCODE and DESCRIPTION) to get only numerical_
↪data
numerical_data = Matrix(data[:, 3:end])

println("Numerical data shape: $(size(numerical_data))")
println("Data type: $(eltype(numerical_data))")

# Check for any missing values
missing_count = sum(ismissing.(numerical_data))
println("Missing values: $missing_count")

# Summary statistics
println("\nData range:")
println("Min value: $(minimum(skipmissing(numerical_data)))")
println("Max value: $(maximum(skipmissing(numerical_data)))")
println("Mean value: $(mean(skipmissing(numerical_data)))")

# Check if data is sparse (many zeros)
zero_count = sum(numerical_data .== 0)
total_elements = length(numerical_data)
```

```
sparsity = zero_count / total_elements
println("Sparsity (fraction of zeros): $(round(sparsity, digits=3))")
```

Numerical data shape: (7444, 37)

Data type: Float64

Missing values: 0

Data range:

Min value: 0.0

Max value: 100.0

Mean value: 0.329505351670854

Sparsity (fraction of zeros): 0.831

This data is carefully constructed and has no missing values. Many values are 0.0, however, which may indicate those measurements are below some threshold.

1.2 2. Creating food categories

We next perform an analysis of this data set that explores general categories of nutrition as supplied in the [What We Eat in America Food Categories](#). We consolidate some of the 15 categories identified there, using instead the following 7 slightly courser categories: - FRUITS: citrus/melons/berries, other fruits, and fruit juice - VEGETABLES: color-based groupings (dark green, red/orange), starchy vegetables (potatoes vs others), legumes, and other vegetables - GRAINS: whole grains and refined grains - PROTEIN FOODS: meat, poultry, seafood (high/low omega-3), eggs, soy, nuts/seeds, and legumes - DAIRY: milk, yogurt, and cheese - FATS & OILS: liquid oils and solid fats - DISCRETIONARY CALORIES: added sugars and alcoholic beverages

An inspection of the column headings above shows the fruit related columns are indexed by F_ terms, vegetables by V_, grains by G_ and so forth. However, there is also a cumulative column F_Total that may be excluded as it is derived from the other columns.

Let us create these categories for our computation.

```
[128]: nutritional_columns = names(data)[3:end] # Skip FOODCODE and DESCRIPTION

# Define natural category groupings based on USDA food patterns
isfruit(col) = startswith(string(col), "F_") && !occursin("TOTAL", string(col))
fruits = filter(isfruit, nutritional_columns)

println("\n FRUITS ($(length(fruits)) categories):")
for (i, fruit) in enumerate(fruits)
    println("  $i. $fruit")
end
```

FRUITS (3 categories):

1. F_CITMLB (cup eq)
2. F_OTHER (cup eq)
3. F_JUICE (cup eq)

```
[129]: isvegetable(col) = startswith(string(col), "V_") && !occursin("TOTAL",
↳string(col))
vegetables = filter(isvegetable, nutritional_columns)
println("\n VEGETABLES ($(length(vegetables)) categories):")
for (i, veg) in enumerate(vegetables)
    println("  $i. $veg")
end
```

```
VEGETABLES (7 categories):
1. V_DRKGR (cup eq)
2. V_REDOR_TOMATO (cup eq)
3. V_REDOR_OTHER (cup eq)
4. V_STARCHY_POTATO (cup eq)
5. V_STARCHY_OTHER (cup eq)
6. V_OTHER (cup eq)
7. V_LEGUMES (cup eq)
```

```
[130]: isgrain(col) = startswith(string(col), "G_") && !occursin("TOTAL", string(col))
grains      = filter(isgrain, nutritional_columns)
println("\n GRAINS ($(length(grains)) categories):")
for (i, grain) in enumerate(grains)
    println("  $i. $grain")
end
```

```
GRAINS (2 categories):
1. G_WHOLE (oz eq)
2. G_REFINED (oz eq)
```

```
[131]: isprotein(col) = startswith(string(col), "PF_") && !occursin("TOTAL",
↳string(col))
proteins = filter(isprotein, nutritional_columns)
println("\n PROTEIN FOODS ($(length(proteins)) categories):")
for (i, protein) in enumerate(proteins)
    println("  $i. $protein")
end
```

```
PROTEIN FOODS (10 categories):
1. PF_MEAT (oz eq)
2. PF_CUREDMEAT (oz eq)
3. PF_ORGAN (oz eq)
4. PF_POULT (oz eq)
5. PF_SEAFD_HI (oz eq)
6. PF_SEAFD_LOW (oz eq)
7. PF_EGGS (oz eq)
8. PF_SOY (oz eq)
9. PF_NUTSDS (oz eq)
```

10. PF_LEGUMES (oz eq)

```
[132]: isdairy(col) = startswith(string(col), "D_") && !occursin("TOTAL", string(col))
dairy      = filter(isdairy, nutritional_columns)
println("\n DAIRY ($(length(dairy)) categories):")
for (i, d) in enumerate(dairy)
    println("  $i. $d")
end
```

DAIRY (3 categories):

1. D_MILK (cup eq)
2. D_YOGURT (cup eq)
3. D_CHEESE (cup eq)

```
[133]: isfat_or_oil(col) = string(col) in ["OILS (grams)", "SOLID_FATS (grams)"]
fats_oils      = filter(isfat_or_oil, nutritional_columns)
println("\n FATS & OILS ($(length(fats_oils)) categories):")
for (i, fat) in enumerate(fats_oils)
    println("  $i. $fat")
end
```

FATS & OILS (2 categories):

1. OILS (grams)
2. SOLID_FATS (grams)
2. SOLID_FATS (grams)

```
[134]: isdiscretionary(col) = string(col) in ["ADD_SUGARS (tsp eq)", "A_DRINKS (no. of_
↳drinks)"]
discretionary = filter(isdiscretionary, nutritional_columns)
println("\n DISCRETIONARY CALORIES ($(length(discretionary)) categories):")
for (i, disc) in enumerate(discretionary)
    println("  $i. $disc")
end
```

DISCRETIONARY CALORIES (2 categories):

1. ADD_SUGARS (tsp eq)
2. A_DRINKS (no. of drinks)

```
[135]: println("SUMMARY OF THE 7 NUTRITIONAL CATEGORIES:")
println("="^50)
println("  FRUITS ($(length(fruits)))")
println("    - Citrus/melons/berries, other fruits, fruit juice (excluding_
↳totals)")

println("\n  VEGETABLES ($(length(vegetables)))")
```



```
println(" - Dark green, red/orange, starchy (potatoes/others), legumes, other_
↳vegetables (excluding totals)")

println("\n GRAINS ($(length(grains))):")
println(" - Whole grains and refined grains (excluding totals)")

println("\n PROTEIN FOODS ($(length(proteins))):")
println(" - Meat, poultry, seafood (high/low omega-3), eggs, soy, nuts/seeds,
↳legumes (excluding totals)")

println("\n DAIRY ($(length(dairy))):")
println(" - Milk, yogurt, cheese (excluding totals)")

println("\n FATS & OILS ($(length(fats_oils))):")
println(" - Liquid oils and solid fats")

println("\n DISCRETIONARY CALORIES ($(length(discretionary))):")
println(" - Added sugars and alcoholic beverages")
```

SUMMARY OF THE 7 NUTRITIONAL CATEGORIES:

=====

FRUITS (3):

- Citrus/melons/berries, other fruits, fruit juice (excluding totals)

VEGETABLES (7):

- Dark green, red/orange, starchy (potatoes/others), legumes, other vegetables (excluding totals)

GRAINS (2):

- Whole grains and refined grains (excluding totals)

PROTEIN FOODS (10):

- Meat, poultry, seafood (high/low omega-3), eggs, soy, nuts/seeds, legumes (excluding totals)

DAIRY (3):

- Milk, yogurt, cheese (excluding totals)

FATS & OILS (2):

- Liquid oils and solid fats

DISCRETIONARY CALORIES (2):

- Added sugars and alcoholic beverages

1.3 3. A “What-We-Eat” tensor

There are many ways to build a tensor from these data. As illustration we explore the relationship of added sugars to products that range over the categories of Fruits, Vegetables, and Proteins.

This produces a 3-way tensor with modes Fruit x Vegetables x Protein, where every entry is the total amount of added sugars for food groups in that category.

Since the data is very sparse, many entries score 0.0 in one or more of the mode categories. In that case we add a special `Empty` class to each of these categories to indicate that the food group does not contain any measureable amount of the specified category. For instance, milk may contain 0.0 detectable Fruit so it would be placed in the `Empty` fruit category.

```
[136]: # Use the specific categories but add "empty" options
fruits_with_empty = vcat(fruits, ["F_EMPTY"])
vegetables_with_empty = vcat(vegetables, ["V_EMPTY"])
proteins_with_empty = vcat(proteins, ["PF_EMPTY"])

println("#Fruits, #Vegetables, #Proteins) = ($(length(fruits_with_empty)),
↪$(length(vegetables_with_empty)), $(length(proteins_with_empty)) )")

(#Fruits, #Vegetables, #Proteins) = (4, 8, 11 )
```

Now we create the tensor with these modes and total added sugar as the value in each entry. We will be using `ITensors` as our backend for tensors. This requires each mode to be created as an `Index` which will prevent the complication of remembering which mode was first, second, or third.

```
[137]: using ITensors

# Create new tensor with empty categories
f = Index(length(fruits_with_empty), "Fruits")
v = Index(length(vegetables_with_empty), "Veg")
p = Index(length(proteins_with_empty), "Prot")

println("\nIndices: $(f) × $(v) × $(p)")
```

```
Indices: (dim=4|id=995|"Fruits") × (dim=8|id=942|"Veg") × (dim=11|id=953|"Prot")
```

To build the tensor we traverse every row of the data and check for a nonzero added sugar value. Upon encountering such a value we then designate the (f,v,p) coordinates of that entry add them to any existing value in the coordinate.

```
[138]: # Rebuild mode indices if they were shadowed by later notebook variables.
if !(f isa Index)
    f = Index(length(fruits_with_empty), "Fruits")
end
if !(v isa Index)
    v = Index(length(vegetables_with_empty), "Veg")
end
if !(p isa Index)
    p = Index(length(proteins_with_empty), "Prot")
end

Sugar = ITensor(f,v,p)
```

```

# Function to explain index meaning
function friendly_label(code::AbstractString)
    label_map = Dict{
        "F_CITMLB (cup eq)" => "Citrus, tomatoes, melons, and berries",
        "F_OTHER (cup eq)" => "Other fruits (apples, bananas, etc.)",
        "F_JUICE (cup eq)" => "Fruit juice",
        "V_DRKGR (cup eq)" => "Dark green vegetables",
        "V_REDOR_TOMATO (cup eq)" => "Red/orange vegetables (tomatoes)",
        "V_REDOR_OTHER (cup eq)" => "Red/orange vegetables (other)",
        "V_STARCHY_POTATO (cup eq)" => "Starchy vegetables (potatoes)",
        "V_STARCHY_OTHER (cup eq)" => "Starchy vegetables (other)",
        "V_OTHER (cup eq)" => "Other vegetables",
        "V_LEGUMES (cup eq)" => "Legumes (as vegetables)",
        "PF_MEAT (oz eq)" => "Meat (beef, pork, lamb, game)",
        "PF_CUREDMEAT (oz eq)" => "Cured meat",
        "PF_ORGAN (oz eq)" => "Organ meats",
        "PF_POULT (oz eq)" => "Poultry",
        "PF_SEAFD_HI (oz eq)" => "High omega-3 seafood",
        "PF_SEAFD_LOW (oz eq)" => "Low omega-3 seafood",
        "PF_EGGS (oz eq)" => "Eggs",
        "PF_SOY (oz eq)" => "Soy products",
        "PF_NUTSDS (oz eq)" => "Nuts and seeds",
        "PF_LEGUMES (oz eq)" => "Legumes (as protein)",
        "F_EMPTY" => "No measurable fruit",
        "V_EMPTY" => "No measurable vegetable",
        "PF_EMPTY" => "No measurable protein"
    }
    return get(label_map, code, code)
end

processed_foods = 0
empty_assignments = Dict{:fruits => 0, :vegetables => 0, :proteins => 0}

for row_idx in 1:nrow(data)
    sugar_value = data[row_idx, "ADD_SUGARS (tsp eq)"]

    if ismissing(sugar_value) || sugar_value == 0.0
        continue
    end

    # Get vector of fruit, veg, and protein for this food entry.
    fruit_vals = [coalesce(data[row_idx, col], 0.0) for col in fruits]
    veg_vals = [coalesce(data[row_idx, col], 0.0) for col in vegetables]
    protein_vals = [coalesce(data[row_idx, col], 0.0) for col in proteins]

    # Determine indices with empty categories

```

```

if maximum(fruit_vals) <= 1e-10
    fruit_idx = length(fruits_with_empty)
    empty_assignments[:fruits] += 1
else
    fruit_idx = argmax(fruit_vals)
end

if maximum(veg_vals) <= 1e-10
    veg_idx = length(vegetables_with_empty)
    empty_assignments[:vegetables] += 1
else
    veg_idx = argmax(veg_vals)
end

if maximum(protein_vals) <= 1e-10
    protein_idx = length(proteins_with_empty)
    empty_assignments[:proteins] += 1
else
    protein_idx = argmax(protein_vals)
end

# Add sugar value to tensor
Sugar[f=>fruit_idx, v=>veg_idx, p=>protein_idx] += sugar_value
processed_foods += 1
end

# Analyze the tensor directly from ITensor entries (no dense Array conversion).
non_zero_empty = 0
max_value = -Inf
for i in 1:ITensors.dim(f), j in 1:ITensors.dim(v), k in 1:ITensors.dim(p)
    entry = Sugar[f=>i, v=>j, p=>k]
    if entry != 0.0
        non_zero_empty += 1
    end
    if entry > max_value
        max_value = entry
    end
end

total_entries = ITensors.dim(f) * ITensors.dim(v) * ITensors.dim(p)

# Print a summary of the tensor.
println("\n Tensor with empty categories created!")
println("Foods with added sugars: $processed_foods, $(round((processed_foods / (
    (size(data,1)-2)) * 100, digits=2)))%")
println("Non-zero entries: $non_zero_empty / $total_entries")

```

```
println("Sparsity: $(round((total_entries - non_zero_empty) / total_entries * 100, digits=1))%")
println("Max value: $(round(max_value, digits=2)) tsp eq")

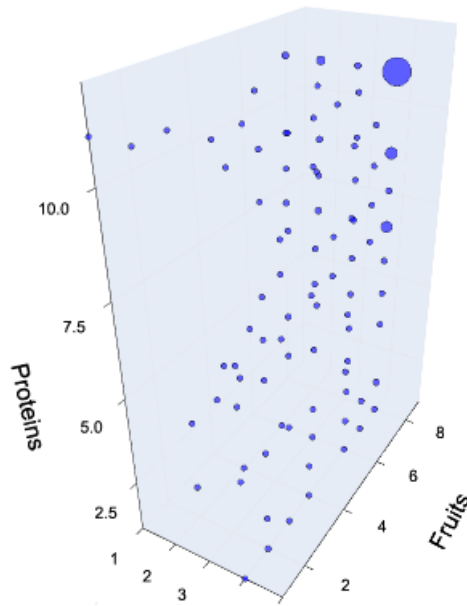
println(" Foods with 0 fruits: $(empty_assignments[:fruits]) foods")
println(" Foods with 0 vegetables: $(empty_assignments[:vegetables]) foods")
println(" Foods with 0 proteins: $(empty_assignments[:proteins]) foods")
```

```
Tensor with empty categories created!
Foods with added sugars: 3194, 42.92%
Non-zero entries: 91 / 352
Sparsity: 74.1%
Max value: 4210.98 tsp eq
  Foods with 0 fruits: 2708 foods
  Foods with 0 vegetables: 2471 foods
  Foods with 0 proteins: 1424 foods
Foods with added sugars: 3194, 42.92%
Non-zero entries: 91 / 352
Sparsity: 74.1%
Max value: 4210.98 tsp eq
  Foods with 0 fruits: 2708 foods
  Foods with 0 vegetables: 2471 foods
  Foods with 0 proteins: 1424 foods
```

```
[139]: inds_sugar = collect(ITensors.inds(Sugar))
plot_tensor(Sugar, title="Sugars Tensor",
            xlabel=mode_label(inds_sugar[2]), ylabel=mode_label(inds_sugar[1]),
            zlabel=mode_label(inds_sugar[3]))
```

[139]:

Sugars Tensor



```
[140]: # Mode-label helpers: use explicit dimension convention first.
# User convention for this notebook: 4=fruit, 8=veg, 11=protein.
function mode_key(ix::Index)
    d = ITensors.dim(ix)
    if d == 4
        return :fruit
    elseif d == 8
        return :veg
    elseif d == 11
        return :protein
    end

    # Fallback for any future tensors that don't follow 4/8/11.
    txt = lowercase(string(tags(ix)) * " " * string(id(ix)))
    if occursin("fruit", txt)
        return :fruit
    elseif occursin("veg", txt) || occursin("vegetable", txt)
        return :veg
    elseif occursin("prot", txt) || occursin("protein", txt)
        return :protein
    else

```

```

        return :unknown
    end
end

function mode_label(ix::Index; mixed=false)
    m = mode_key(ix)
    if m == :fruit
        return mixed ? "Fruit mix" : "Fruits"
    elseif m == :veg
        return mixed ? "Vegetable mix" : "Vegetables"
    elseif m == :protein
        return mixed ? "Protein mix" : "Proteins"
    else
        return mixed ? "Mixed mode" : "Mode"
    end
end
end

```

[140]: mode_label (generic function with 1 method)

This data shows that 42% of the foods consumed in the USA for this study contained added sugars. We can see this some what sparse and may have some clustering but it is spread out and perhaps just an optical illusion. We will tart to look at the tensor with existing tools then turn to Dleto's stratify operations to see if we can detect the types of products that use the most added sugars.

1.4 3.1 Sample the Tensor

For instance, our tensors 1,1,1 entry is the total added sugars for food products that have 1 serving of fruit X, Protein Y, and Vegetable Z.

```

[141]: # Human-readable interpretation of the (1,1,1) sugar tensor entry
fruit_code = fruits_with_empty[1]
veg_code = vegetables_with_empty[1]
protein_code = proteins_with_empty[1]

label_map = Dict(
    "F_CITMLB (cup eq)" => "Citrus, tomatoes, melons, and berries",
    "V_DRKGR (cup eq)" => "Dark green vegetables",
    "PF_MEAT (oz eq)" => "Meat (beef, pork, lamb, game)"
)

fruit_label = get(label_map, fruit_code, fruit_code)
veg_label = get(label_map, veg_code, veg_code)
protein_label = get(label_map, protein_code, protein_code)

sugar_tsp = Sugar[f=>1, v=>1, p=>1]
println("(1,1,1) corresponds to: $fruit_label + $veg_label + $protein_label")
println("Total added sugar in this bucket: $(round(sugar_tsp, digits=2)) tsp_
↳eq")

```

(1,1,1) corresponds to: Citrus, tomatoes, melons, and berries + Dark green vegetables + Meat (beef, pork, lamb, game)
 Total added sugar in this bucket: 0.0 tsp eq

Let us find an entry that is nonzero

```
[142]: idx = nothing
val = 0.0

for i in 1:dim(f), j in 1:dim(v), k in 1:dim(p)
  entry = Sugar[f=>i, v=>j, p=>k]
  if entry > 0
    idx = (i, j, k)
    val = entry
    break
  end
end

if isnothing(idx)
  println("No nonzero entries found in Sugar.")
else
  i, j, k = idx

  fruit_code = fruits_with_empty[i]
  veg_code = vegetables_with_empty[j]
  protein_code = proteins_with_empty[k]

  println("Found nonzero entry at (i,j,k) = ($i,$j,$k)")
  println("Meaning: $(friendly_label(fruit_code)) +␣
  ↪$(friendly_label(veg_code)) + $(friendly_label(protein_code))")
  println("Total added sugar in this bucket: $(round(val, digits=2)) tsp eq")
end
```

Found nonzero entry at (i,j,k) = (1,1,11)
 Meaning: Citrus, tomatoes, melons, and berries + Dark green vegetables + No measurable protein
 Total added sugar in this bucket: 0.4 tsp eq

1.5 3.2 Look for Tucker decompositions.

Now let us use Dleto. We start by asking considering a Tucker decomposition, also known as a degeneracy. In the original data the Tucker decompositions are all trivial up the default tolerances.

```
[143]: # Apply Dleto tensor analysis: nondeg and stratify
using Dleto

nondeg_Sugar, Ys = nondeg(Sugar)
# println("\nNondegenerate tensor created with $(nondeg_Sugar) non-zero␣
  ↪entries")
```



```

# Compare original index dimensions to nondegenerate index dimensions
↳ (order-invariant).
orig_inds = collect(ITensors.inds(Sugar))
nondeg_inds = collect(ITensors.inds(nondeg_Sugar))

orig_dims = [ITensors.dim(i) for i in orig_inds]
nondeg_dims = [ITensors.dim(i) for i in nondeg_inds]

sorted_orig_dims = sort(orig_dims)
sorted_nondeg_dims = sort(nondeg_dims)
dim_diffs = sorted_orig_dims .- sorted_nondeg_dims

println("Sorted original dims: $sorted_orig_dims")
println("Sorted nondeg dims:   $sorted_nondeg_dims")
println("Dimension differences after sorting (orig - nondeg): $dim_diffs")

```

Sorted original dims: [4, 8, 11]

Sorted nondeg dims: [4, 8, 11]

Dimension differences after sorting (orig - nondeg): [0, 0, 0]

Tucker decompositions were not informative.

1.6 3.3 Look for a HOSVD structure.

Next we look to see what is signaled by the higher-order singular values.

```

[144]: using LinearAlgebra

# ITensor-native HOSVD on nondeg_Sugar (no dense Array conversion).
T_hosvd = nondeg_Sugar
mode_inds = collect(ITensors.inds(T_hosvd))
N = length(mode_inds)

U_factors = Vector{ITensor}(undef, N)
link_inds = Vector{Index}(undef, N)
svs = Vector{Vector{Float64}}(undef, N)

for n in 1:N
    U, S, V = ITensors.svd(T_hosvd, mode_inds[n])
    U_factors[n] = U

    s1, s2 = ITensors.inds(S)
    link_inds[n] = s1
    svs[n] = [S[s1=>k, s2=>k] for k in 1:ITensors.dim(s1)]
end

# Core tensor G = T ×1 U1' ×2 U2' ... ×N UN'

```

```

G_it = T_hosvd
for n in 1:N
    G_it *= dag(U_factors[n])
end

# Reconstruct from HOSVD and report relative error.
A_hat_it = G_it
for n in 1:N
    A_hat_it *= U_factors[n]
end
rel_err = norm(T_hosvd - A_hat_it) / max(norm(T_hosvd), eps(Float64))

orig_dims = [ITensors.dim(i) for i in mode_inds]
core_dims = [ITensors.dim(i) for i in ITensors.inds(G_it)]
factor_dims = [(ITensors.dim(mode_inds[n]), ITensors.dim(link_inds[n])) for n in 1:N]

println("HOSVD computed for order- $N$  tensor.")
println("Original dims: $orig_dims")
println("Core dims:      $core_dims")
println("Factor dims (mode, rank): $factor_dims")
println("Relative reconstruction error: $(rel_err)")

```

```

HOSVD computed for order-3 tensor.
Original dims: [11, 8, 4]
Core dims:      [11, 8, 4]
Factor dims (mode, rank): [(11, 11), (8, 8), (4, 4)]
Relative reconstruction error: 9.526529123620498e-16
Original dims: [11, 8, 4]
Core dims:      [11, 8, 4]
Factor dims (mode, rank): [(11, 11), (8, 8), (4, 4)]
Relative reconstruction error: 9.526529123620498e-16

```

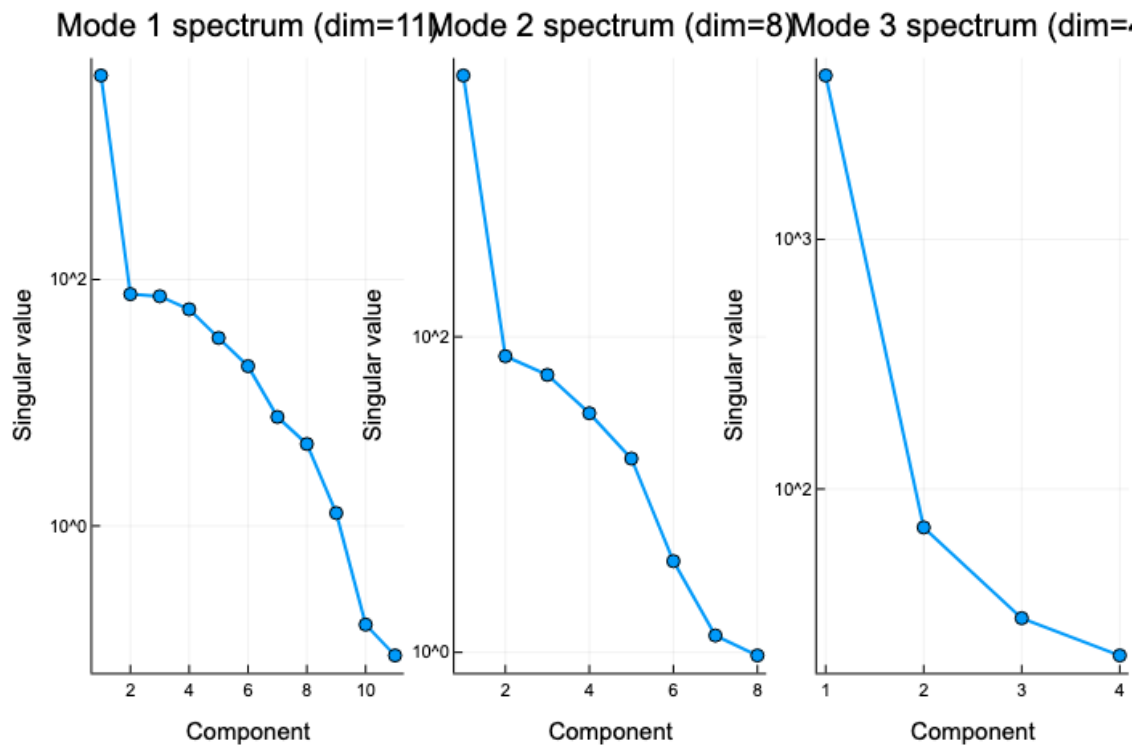
Plot the higher order singular values in each mode and then the core tensor of the highest value.

```

[145]: # Plot HOSVD spectra from ITensor singular values (mode unfoldings).
p = plot(layout=(1, N), size=(420 * N, 320), legend=false)
for n in 1:N
    plot!(
        p[n],
        1:length(svs[n]),
        max.(svs[n], eps(Float64)),
        marker=:circle,
        linewidth=2,
        xlabel="Component",
        ylabel="Singular value",
        yscale=:log10,
        title="Mode $n spectrum (dim=$(orig_dims[n]))"
    )
end

```

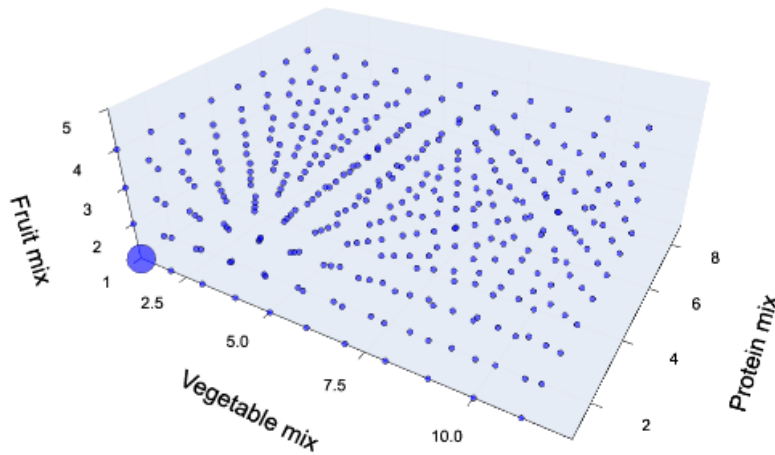
```
)
end
display(p)
```



```
[146]: # Plot the tensor transformed by HOSVD (the core tensor G) with labels from
        ↪ index names.
inds_core = collect(ITensors.inds(G_it))
plot_tensor(
    G_it,
    title="HOSVD Core Tensor",
    xlabel=mode_label(inds_core[2]; mixed=true),
    ylabel=mode_label(inds_core[1]; mixed=true),
    zlabel=mode_label(inds_core[3]; mixed=true)
)
```

[146]:

HOSVD Core Tensor



1.7 3.4 ParaFAC decomposition.

Last we look for a feature in low-rank approximation techniques, specifically ParaFAC.

```
[147]: using Random, LinearAlgebra

# PARAFAC / CP-ALS decomposition of the sugar tensor.
# We use a rank-R model and reconstruct the approximating tensor for display.
@isdefined(Sugar) || error("Sugar is not defined. Run the sugar tensor_
↳construction cells first.")

function khatri_rao(A::AbstractMatrix, B::AbstractMatrix)
    size(A, 2) == size(B, 2) || error("Khatri-Rao requires same number of_
↳columns.")
    m, r = size(A)
    n, _ = size(B)
    KR = zeros(promote_type(eltype(A), eltype(B)), m * n, r)
    for j in 1:r
        # Ordering matches unfold3 definitions below.
        KR[:, j] = vec(kron(A[:, j], B[:, j]))
    end
    return KR
end
```

```

end

function unfold3(X::Array{Float64,3}, mode::Int)
    if mode == 1
        return reshape(permutedims(X, (1, 2, 3)), size(X, 1), :)
    elseif mode == 2
        return reshape(permutedims(X, (2, 1, 3)), size(X, 2), :)
    elseif mode == 3
        return reshape(permutedims(X, (3, 1, 2)), size(X, 3), :)
    else
        error("mode must be 1, 2, or 3")
    end
end

function cp_reconstruct(A::Matrix{Float64}, B::Matrix{Float64}, C::
    ↪Matrix{Float64}, lambda::Vector{Float64})
    I1, R = size(A)
    J1, _ = size(B)
    K1, _ = size(C)
    Xhat = zeros(Float64, I1, J1, K1)
    for r in 1:R
        Xhat .+= lambda[r] .* reshape(A[:, r], I1, 1, 1) .* reshape(B[:, r], 1, ↪
    ↪J1, 1) .* reshape(C[:, r], 1, 1, K1)
    end
    return Xhat
end

function cp_als_3way(X::Array{Float64,3}, R::Int; maxiter::Int=200, tol::
    ↪Float64=1e-7, seed::Int=1234, ridge::Float64=1e-8)
    I1, J1, K1 = size(X)
    Random.seed!(seed)

    # Positive initialization usually behaves better for nonnegative survey ↪
    ↪tensors.
    A = rand(I1, R)
    B = rand(J1, R)
    C = rand(K1, R)
    lambda = ones(Float64, R)

    X1 = unfold3(X, 1)
    X2 = unfold3(X, 2)
    X3 = unfold3(X, 3)

    prev_relerr = Inf
    relerr = Inf

    for _ in 1:maxiter

```

```

        KRcb = khatrib_rao(C, B)
        G1 = (B' * B) .* (C' * C) + ridge * Matrix{Float64}(LinearAlgebra.I, R, R)
    end
    A = (X1 * KRcb) / G1

    KRca = khatrib_rao(C, A)
    G2 = (A' * A) .* (C' * C) + ridge * Matrix{Float64}(LinearAlgebra.I, R, R)
    B = (X2 * KRca) / G2

    KRba = khatrib_rao(B, A)
    G3 = (A' * A) .* (B' * B) + ridge * Matrix{Float64}(LinearAlgebra.I, R, R)
    C = (X3 * KRba) / G3

    # Normalize columns each iteration and absorb scales in lambda.
    lambda .= 1.0
    for r in 1:R
        nA = norm(@view A[:, r]); nA = nA > 0 ? nA : 1.0
        nB = norm(@view B[:, r]); nB = nB > 0 ? nB : 1.0
        nC = norm(@view C[:, r]); nC = nC > 0 ? nC : 1.0
        A[:, r] ./= nA
        B[:, r] ./= nB
        C[:, r] ./= nC
        lambda[r] = nA * nB * nC
    end

    Xhat = cp_reconstruct(A, B, C, lambda)
    relerr = norm(X - Xhat) / max(norm(X), eps{Float64})

    if abs(prev_relerr - relerr) < tol
        break
    end
    prev_relerr = relerr
end

return A, B, C, lambda, relerr
end

# Build a dense copy for CP-ALS, then map the reconstruction back to ITensor
for plotting.
sugar_inds = collect(ITensors.inds(Sugar))
I = ITensors.dim(sugar_inds[1])
J = ITensors.dim(sugar_inds[2])
K = ITensors.dim(sugar_inds[3])

X_sugar = zeros{Float64, I, J, K}

```

```

for i in 1:I, j in 1:J, k in 1:K
    X_sugar[i, j, k] = Sugar[sugar_inds[1]=>i, sugar_inds[2]=>j,
    ↪sugar_inds[3]=>k]
end

cp_rank = 5
A_cp, B_cp, C_cp, lambda_cp, relerr_cp = cp_als_3way(X_sugar, cp_rank;
    ↪maxiter=300, tol=1e-8, seed=42, ridge=1e-7)
Xhat_cp = cp_reconstruct(A_cp, B_cp, C_cp, lambda_cp)

Sugar_parafac = ITensor(sugar_inds...)
for i in 1:I, j in 1:J, k in 1:K
    Sugar_parafac[sugar_inds[1]=>i, sugar_inds[2]=>j, sugar_inds[3]=>k] =
    ↪Xhat_cp[i, j, k]
end

println("PARAFAC rank: $cp_rank")
println("PARAFAC relative reconstruction error: $(round(relerr_cp, digits=6))")

plot_tensor(
    Sugar_parafac,
    title="Sugar Tensor PARAFAC Approximation (rank=$cp_rank)",
    xlabel=mode_label(sugar_inds[2]),
    ylabel=mode_label(sugar_inds[1]),
    zlabel=mode_label(sugar_inds[3])
)

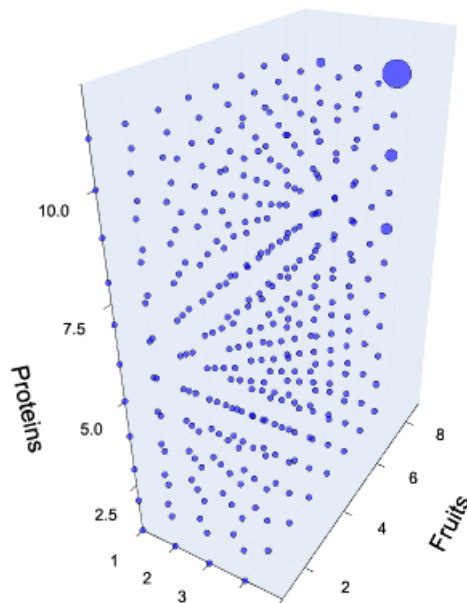
```

PARAFAC rank: 5

PARAFAC relative reconstruction error: 0.005782

[147]:

Sugar Tensor PARAFAC Approximation (rank=5)



This tensor is now appears dense and no reduction in dimension or clusters seem to have appeared.

2 Stratification

We now run Dleto chiseling algorithms. First with the universal chisel, then with a hint at what we find an adjoint chisel. By rotating the image one can identify the generic stratification algorithm has uncovered a cluster that which is 1 x 1 in the

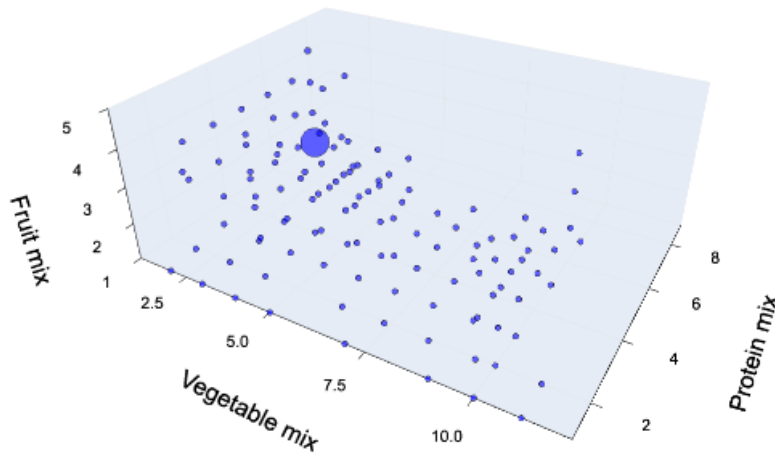
```
[148]: @time Sugar_strat, Xs_strat = stratify(nondeg_Sugar)
inds_strat = collect(ITensors.inds(Sugar_strat))
plot_tensor(
    Sugar_strat,
    title="Dleto Stratified Tensor",
    xlabel=mode_label(inds_strat[2]; mixed=true),
    ylabel=mode_label(inds_strat[1]; mixed=true),
    zlabel=mode_label(inds_strat[3]; mixed=true)
)
```

[Info: Found 7 derivations for stratification.

0.041214 seconds (284.39 k allocations: 36.677 MiB, 38.72% gc time)

[148]:

Dieto Stratified Tensor



We see in fact that there 7 derivations. For a 3 tensor 2 are always trivial so we have 5 possible independent derivations to explore features. The first displayed already we can look at the others.

```
[149]: # Explore derivation vectors ivec = 3:7 one by one and store results for reuse.
ch = UniversalChisel(length(ITensors.inds(nondeg_Sugar)))
fr = collect(ITensors.inds(nondeg_Sugar))
Ω = IndTransverseOps(fr, UniversalOp())

Sugar_strat_by_ivec = Dict{Int, ITensor}()
Xs_strat_by_ivec = Dict{Int, Vector{ITensor}}()

for ivec in 3:7
    println("\n--- Stratification with ivec = $ivec ---")
    Sugar_strat_i, Xs_strat_i = stratify(Ω, ch, nondeg_Sugar; ivec=ivec)

    # Store outputs so later cells can access them by ivec key.
    Sugar_strat_by_ivec[ivec] = Sugar_strat_i
    Xs_strat_by_ivec[ivec] = Xs_strat_i

    inds_i = collect(ITensors.inds(Sugar_strat_i))
    display(
        plot_tensor(
```

```

        Sugar_strat_i,
        title="Stratified Tensor (ivec=$ivec)",
        xlabel=mode_label(inds_i[1]; mixed=true),
        ylabel=mode_label(inds_i[2]; mixed=true),
        zlabel=mode_label(inds_i[3]; mixed=true)
    )
end

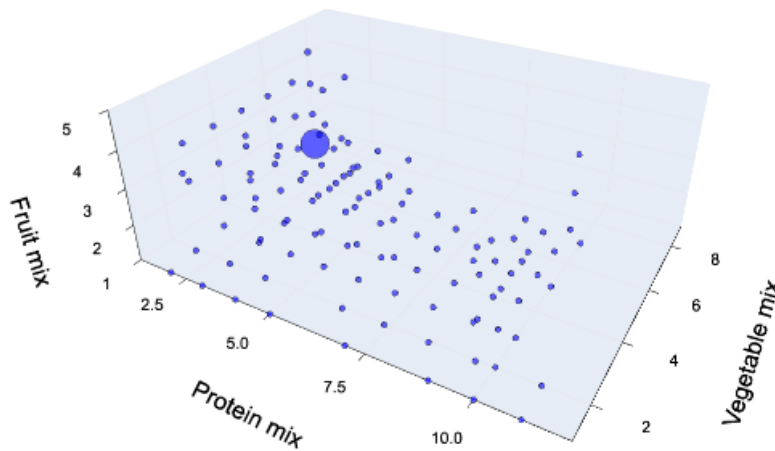
println("Stored ivec keys: $(collect(keys(Sugar_strat_by_ivec)))")

```

--- Stratification with ivec = 3 ---

[Info: Found 7 derivations for stratification.

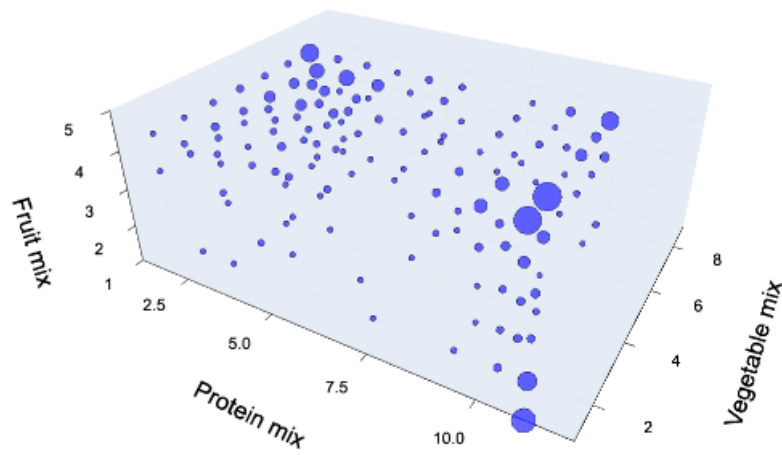
Stratified Tensor (ivec=3)



--- Stratification with ivec = 4 ---

[Info: Found 7 derivations for stratification.

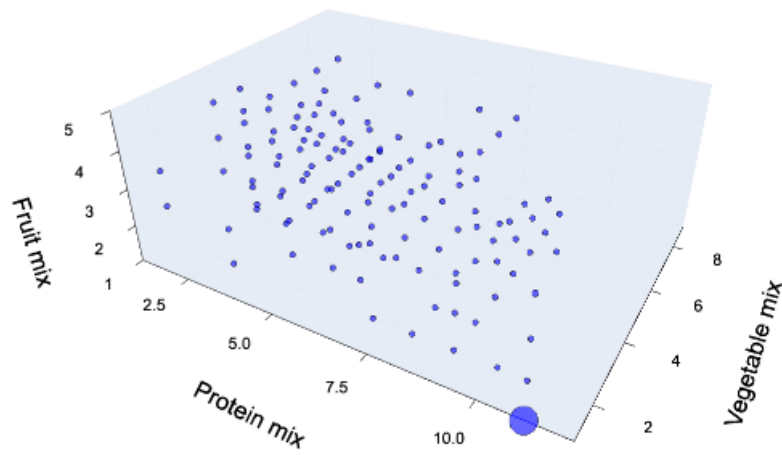
Stratified Tensor (ivec=4)



--- Stratification with ivec = 5 ---

[Info: Found 7 derivations for stratification.

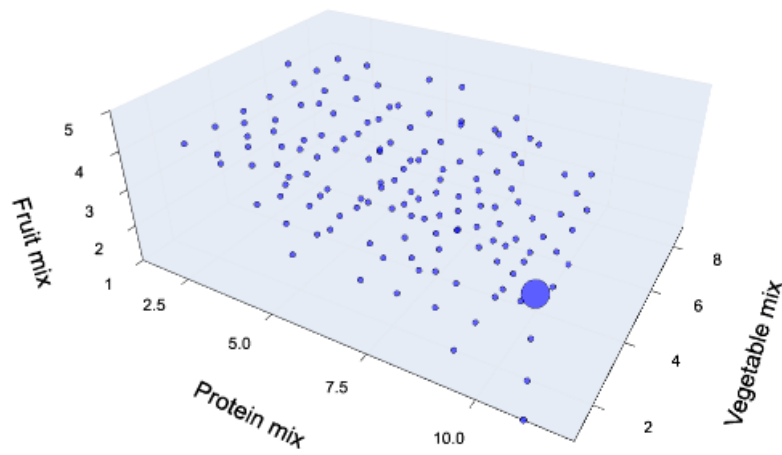
Stratified Tensor (ivec=5)



--- Stratification with ivec = 6 ---

[Info: Found 7 derivations for stratification.

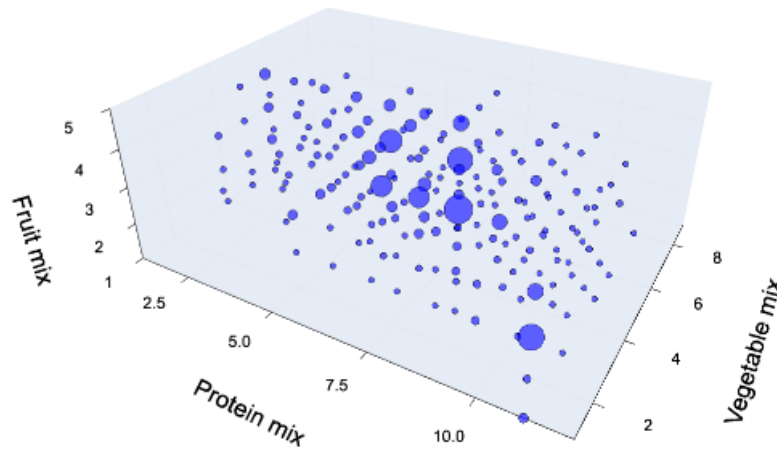
Stratified Tensor (ivec=6)



--- Stratification with iverc = 7 ---

[Info: Found 7 derivations for stratification.

Stratified Tensor (ivec=7)



Stored ivec keys: [5, 4, 6, 7, 3]

Using `ivec=3` as the one to explore, i.e. the original returned by `stratify`, we can convert the basis of the outlier block back into the mixture of fruits and vegetables the produced it. Seeing that the clustering is spread along the protein axis there is no relevance to that.

[]:

Let us now print out the mixtures. Remember this is an analytic probe so we allow negative coefficients.

```
[150]: # Pivoted tables for ivec=3..7 using fixed cluster coordinates.
# Output focused on protein and vegetable modes (fruit omitted from reporting).
ivec_list = collect(3:7)

function fixed_cluster_coord_and_value(T::ITensor, iv::Int)
    inds_mix = collect(ITensors.inds(T))
    dims_mix = [ITensors.dim(i) for i in inds_mix]

    # Locate positions by dimension convention: 4=fruit, 8=veg, 11=protein.
    pos_fruit = findfirst(==(4), dims_mix)
    pos_veg = findfirst(==(8), dims_mix)
    pos_prot = findfirst(==(11), dims_mix)
```

```

isnothing(pos_fruit) && error("Could not locate fruit mode (dim=4).")
isnothing(pos_veg) && error("Could not locate vegetable mode (dim=8).")
isnothing(pos_prot) && error("Could not locate protein mode (dim=11).")

req_p, req_v = iv == 3 ? (1, 9) : (11, 1)
prot_idx = clamp(req_p, 1, dims_mix[pos_prot])
veg_idx = clamp(req_v, 1, dims_mix[pos_veg])

# Pick fruit coordinate by largest magnitude on the fixed (p,v) slice.
best_fruit = 1
best_val = -Inf
for fi in 1:dims_mix[pos_fruit]
    coord = [1, 1, 1]
    coord[pos_fruit] = fi
    coord[pos_veg] = veg_idx
    coord[pos_prot] = prot_idx
    vabs = abs(T[inds_mix[1]=>coord[1], inds_mix[2]=>coord[2],
↳inds_mix[3]=>coord[3]])
    if vabs > best_val
        best_val = vabs
        best_fruit = fi
    end
end

best_coord = [1, 1, 1]
best_coord[pos_fruit] = best_fruit
best_coord[pos_veg] = veg_idx
best_coord[pos_prot] = prot_idx

return inds_mix, dims_mix, Tuple(best_coord), best_val, req_p, req_v,
↳prot_idx, veg_idx
end

function print_pivot(categories::Vector{String}, value_by_cat::
↳Dict{String,Dict{Int,Float64}}, ivecs::Vector{Int};
↳value_fmt=x->string(round(x, digits=1), "%")
    headers = vcat(["category"], ["ivec=$iv" for iv in ivecs])

    rows = Vector{Vector{String}}()
    for cat in categories
        row = String[cat]
        for iv in ivecs
            v = get(get(value_by_cat, cat, Dict{Int,Float64}{}), iv, NaN)
            push!(row, isnan(v) ? "" : value_fmt(v))
        end
        push!(rows, row)
    end
end

```

```

widths = [length(h) for h in headers]
for row in rows
    for j in eachindex(row)
        widths[j] = max(widths[j], length(row[j]))
    end
end

println(join([rpad(headers[j], widths[j]) for j in eachindex(headers)], " | "
↳"))
println(join([repeat("-", w) for w in widths], "-+-"))
for row in rows
    println(join([rpad(row[j], widths[j]) for j in eachindex(row)], " | "))
end
end

summary_abs = Dict{Int,Float64}()
summary_pv = Dict{Int,String}()
protein_by_cat = Dict{String,Dict{Int,Float64}}()
veg_by_cat = Dict{String,Dict{Int,Float64}}()

protein_order = [friendly_label(name) for name in proteins_with_empty]
veg_order = [friendly_label(name) for name in vegetables_with_empty]

for iv in ivec_list
    haskey(Sugar_strat_by_ivec, iv) || error("Missing Sugar_strat_by_ivec[$iv]. "
↳Run the ivec storage cell.")
    haskey(Xs_strat_by_ivec, iv) || error("Missing Xs_strat_by_ivec[$iv]. Run "
↳the ivec storage cell.")

    Sugar_strat_iv = Sugar_strat_by_ivec[iv]
    Xs_iv = Xs_strat_by_ivec[iv]

    inds_mix, dims_mix, best_coord, best_val, req_p, req_v, prot_idx, veg_idx =
↳fixed_cluster_coord_and_value(Sugar_strat_iv, iv)
    summary_abs[iv] = best_val
    summary_pv[iv] = "(p=>$prot_idx, v=>$veg_idx)"

    pos_prot = findfirst(==(11), dims_mix)
    pos_veg = findfirst(==(8), dims_mix)

    X_prot = first(X for X in Xs_iv if any(ITensors.dim(ix) == 11 for ix in
↳ITensors.inds(X)))
    X_veg = first(X for X in Xs_iv if any(ITensors.dim(ix) == 8 for ix in
↳ITensors.inds(X)))

```



```

coord_prot = best_coord[pos_prot]
coord_veg = best_coord[pos_veg]

mix_prot_idx = inds_mix[pos_prot]
mix_veg_idx = inds_mix[pos_veg]

e_prot = ITensor(mix_prot_idx); e_prot[mix_prot_idx=>coord_prot] = 1.0
e_veg = ITensor(mix_veg_idx); e_veg[mix_veg_idx=>coord_veg] = 1.0

prot_mix_old = dag(X_prot) * e_prot
veg_mix_old = dag(X_veg) * e_veg

prot_old_idx = only([ix for ix in ITensors.inds(prot_mix_old) if ITensors.
↳dim(ix) == 11])
veg_old_idx = only([ix for ix in ITensors.inds(veg_mix_old) if ITensors.
↳dim(ix) == 8])

prot_weights = [prot_mix_old[prot_old_idx=>i] for i in 1:ITensors.
↳dim(prot_old_idx)]
veg_weights = [veg_mix_old[veg_old_idx=>i] for i in 1:ITensors.
↳dim(veg_old_idx)]

for (name, w) in zip(proteins_with_empty, prot_weights)
    abs(w) > 1e-3 || continue
    cat = friendly_label(name)
    get!(protein_by_cat, cat, Dict{Int,Float64}{}())[iv] = 100 * w
end

for (name, w) in zip(vegetables_with_empty, veg_weights)
    abs(w) > 1e-3 || continue
    cat = friendly_label(name)
    get!(veg_by_cat, cat, Dict{Int,Float64}{}())[iv] = 100 * w
end
end

println("\nFixed Cluster Summary (protein/veg only)")
coord_rows = [
    ["selected (p,v)"; [summary_pv[iv] for iv in ivec_list]...],
    ["|value|"; [string(round(summary_abs[iv], digits=4)) for iv in ivec_list]..
↳.]
]
]
coord_headers = vcat(["metric"], ["ivec=$iv" for iv in ivec_list])
coord_widths = [length(h) for h in coord_headers]
for row in coord_rows
    for j in eachindex(row)
        coord_widths[j] = max(coord_widths[j], length(row[j]))
    end
end

```

```

end
println(join([rpad(coord_headers[j], coord_widths[j]) for j in_
↳eachindex(coord_headers)], " | "))
println(join([repeat("-", w) for w in coord_widths], "--"))
for row in coord_rows
    println(join([rpad(row[j], coord_widths[j]) for j in eachindex(row)], " |_
↳"))
end

println("\nProtein Mixtures (%; ivec in columns)")
protein_categories = [c for c in protein_order if haskey(protein_by_cat, c)]
print_pivot(protein_categories, protein_by_cat, ivec_list)

println("\nVegetable Mixtures (%; ivec in columns)")
veg_categories = [c for c in veg_order if haskey(veg_by_cat, c)]
print_pivot(veg_categories, veg_by_cat, ivec_list)

```

Fixed Cluster Summary (protein/veg only)

metric	ivec=3	ivec=4	ivec=5	ivec=6	ivec=7
selected (p,v)	(p=>1, v=>8)	(p=>11, v=>1)	(p=>11, v=>1)	(p=>11, v=>1)	(p=>11, v=>1)
value	0.0757	0.0022	1.0629	1.128	0.0049

Protein Mixtures (%; ivec in columns)

category	ivec=3	ivec=4	ivec=5	ivec=6	ivec=7
Meat (beef, pork, lamb, game)	36.1%	16.3%	-15.2%	-5.5%	-19.0%
Cured meat	15.7%	-14.5%	-9.4%	11.9%	13.6%
Organ meats	4.3%	1.1%	-75.9%	-84.6%	1.1%
Poultry	38.3%	10.1%	-17.3%	-1.4%	-12.1%
High omega-3 seafood	1.7%	-1.7%	-19.3%	-19.5%	2.1%
Low omega-3 seafood	12.6%	-7.0%	-7.4%	5.9%	5.6%
Eggs	65.2%	50.4%	-21.3%	-20.0%	-53.9%
Soy products	-0.8%	2.0%	-9.6%	-13.2%	-1.4%
Nuts and seeds	40.2%	-74.4%	-44.4%	37.3%	71.9%
Legumes (as protein)	-13.3%	34.7%	18.6%	-16.0%	-33.9%
No measurable protein	-27.4%	8.5%	17.9%	-5.4%	-7.0%

Vegetable Mixtures (%; ivec in columns)

category	ivec=3	ivec=4	ivec=5	ivec=6	ivec=7
Dark green vegetables	74.5%	74.5%	-74.5%	74.5%	-74.5%

Red/orange vegetables (tomatoes)	0.3%	0.3%	-0.3%	0.3%	
Red/orange vegetables (other)	-1.7%	-1.7%	1.7%	-1.7%	1.6%
Starchy vegetables (potatoes)	-53.7%	-53.7%	53.7%	-53.7%	53.7%
Starchy vegetables (other)	39.5%	39.5%	-39.5%	39.5%	-39.5%
Other vegetables	-1.1%	-1.1%	1.1%	-1.1%	0.5%
Legumes (as vegetables)	0.4%	0.4%	-0.4%	0.4%	-0.5%
No measurable vegetable					0.8%

3 Adjoint Chisel Stratification

Since we saw the most obvious clustering in the fruit-vegetable axes we next design a chisel to focus only on these. This is a version of what is known as an *Adjoint Chisel*.

```
[151]: # Use an adjoint chisel on the fruit/vegetable axes so the protein axis stays
        ↪ at 0.
ch = AdjointChisel(3, 1, 2)
fr = collect(ITensors.inds(nondeg_Sugar))
Ω = IndTransverseOps(fr, UniversalOp())

@time Sugar_adj_strat, Xs_strat = stratify(Ω, ch, nondeg_Sugar; tol=1e-6);
```

0.019606 seconds (216.06 k allocations: 24.763 MiB)

[Info: Found 4 derivations for stratification.

```
[152]: # Plot the nontrivial adjoint-chisel derivations ivec = 2, 3, 4,
        # then print protein/vegetable vector tables for requested fixed clusters.
ch_adj = AdjointChisel(3, 1, 2)
fr_adj = collect(ITensors.inds(nondeg_Sugar))
Ω_adj = IndTransverseOps(fr_adj, UniversalOp())

adj_by_ivec = Dict{Int,ITensor}()
adj_Xs_by_ivec = Dict{Int,Vector{ITensor}}()

for ivec in 2:4
    println("\n--- Adjoint stratification with ivec = $ivec ---")
    Sugar_adj_i, Xs_adj_i = stratify(Ω_adj, ch_adj, nondeg_Sugar; tol=1e-6,
        ↪ ivec=ivec)
    adj_by_ivec[ivec] = Sugar_adj_i
    adj_Xs_by_ivec[ivec] = Xs_adj_i

    inds_adj_i = collect(ITensors.inds(Sugar_adj_i))
    display(
        plot_tensor(
            Sugar_adj_i,
            title="Adjoint Stratified Tensor (ivec=$ivec)",
            xlabel=mode_label(inds_adj_i[1]; mixed=true),
```

```

        ylabel=mode_label(inds_adj_i[2]; mixed=true),
        xlabel=mode_label(inds_adj_i[3]; mixed=true)
    )
end

# Fixed cluster coordinates requested by user:
# ivec=2 and 4: (p=>11, v=>8), ivec=3: (p=>1, v=>1)
fixed_req = Dict{2 => (11, 8), 3 => (1, 1), 4 => (11, 8)}

function print_pivot(categories::Vector{String}, value_by_cat::
    Dict{String,Dict{Int,Float64}}, ivecs::Vector{Int};
    value_fmt=x->string(round(x, digits=1), "%"))
    headers = vcat(["category"], ["ivec=$iv" for iv in ivecs])

    rows = Vector{Vector{String}}()
    for cat in categories
        row = String[cat]
        for iv in ivecs
            v = get(get(value_by_cat, cat, Dict{Int,Float64}{}), iv, NaN)
            push!(row, isnan(v) ? "" : value_fmt(v))
        end
        push!(rows, row)
    end

    widths = [length(h) for h in headers]
    for row in rows
        for j in eachindex(row)
            widths[j] = max(widths[j], length(row[j]))
        end
    end

    println(join([rpad(headers[j], widths[j]) for j in eachindex(headers)], " | "
    ))
    println(join([repeat("-", w) for w in widths], "-+-"))
    for row in rows
        println(join([rpad(row[j], widths[j]) for j in eachindex(row)], " | "))
    end
end

ivec_list = [2, 3, 4]
protein_by_cat = Dict{String,Dict{Int,Float64}}{}
veg_by_cat = Dict{String,Dict{Int,Float64}}{}
protein_order = [friendly_label(name) for name in proteins_with_empty]
veg_order = [friendly_label(name) for name in vegetables_with_empty]
summary_rows = NamedTuple[]

```

```

for ivec in ivec_list
    T = adj_by_ivec[ivec]
    Xs = adj_Xs_by_ivec[ivec]
    inds_mix = collect(ITensors.inds(T))
    dims_mix = [ITensors.dim(i) for i in inds_mix]

    pos_veg = findfirst(==(8), dims_mix)
    pos_prot = findfirst(==(11), dims_mix)
    isnothing(pos_veg) && error("Could not locate vegetable mode (dim=8) for_
↳ivec=$ivec.")
    isnothing(pos_prot) && error("Could not locate protein mode (dim=11) for_
↳ivec=$ivec.")

    req_p, req_v = fixed_req[ivec]
    prot_idx = clamp(req_p, 1, dims_mix[pos_prot])
    veg_idx = clamp(req_v, 1, dims_mix[pos_veg])

    X_prot = first(X for X in Xs if any(ITensors.dim(ix) == 11 for ix in_
↳ITensors.inds(X)))
    X_veg = first(X for X in Xs if any(ITensors.dim(ix) == 8 for ix in ITensors.
↳inds(X)))

    mix_prot_idx = inds_mix[pos_prot]
    mix_veg_idx = inds_mix[pos_veg]

    # Use fixed requested (p,v) coordinates directly for vector reconstruction.
    e_prot = ITensor(mix_prot_idx); e_prot[mix_prot_idx=>prot_idx] = 1.0
    e_veg = ITensor(mix_veg_idx); e_veg[mix_veg_idx=>veg_idx] = 1.0

    prot_mix_old = dag(X_prot) * e_prot
    veg_mix_old = dag(X_veg) * e_veg

    prot_old_idx = only([ix for ix in ITensors.inds(prot_mix_old) if ITensors.
↳dim(ix) == 11])
    veg_old_idx = only([ix for ix in ITensors.inds(veg_mix_old) if ITensors.
↳dim(ix) == 8])

    prot_weights = [prot_mix_old[prot_old_idx=>i] for i in 1:ITensors.
↳dim(prot_old_idx)]
    veg_weights = [veg_mix_old[veg_old_idx=>i] for i in 1:ITensors.
↳dim(veg_old_idx)]

    for (name, w) in zip(proteins_with_empty, prot_weights)
        abs(w) > 1e-3 || continue
        cat = friendly_label(name)
        get!(protein_by_cat, cat, Dict{Int,Float64}{}())[ivec] = 100 * w

```

```

end

for (name, w) in zip(vegetables_with_empty, veg_weights)
  abs(w) > 1e-3 || continue
  cat = friendly_label(name)
  get!(veg_by_cat, cat, Dict{Int,Float64}{})[ivec] = 100 * w
end

push!(summary_rows, (
  ivec = ivec,
  requested = "(p=>$req_p, v=>$req_v)",
  used = "(p=>$prot_idx, v=>$veg_idx)"
))
end

# Print summary + vector tables after the three 3D plots.
headers = ["ivec", "requested", "used"]
getters = [r -> r.ivec, r -> r.requested, r -> r.used]
widths = [length(h) for h in headers]
for row in summary_rows
  for (j, g) in enumerate(getters)
    widths[j] = max(widths[j], length(string(g(row))))
  end
end

println("\nFixed Cluster Table (Adjoint ivec=2,3,4)")
println(join([rpad(headers[j], widths[j]) for j in eachindex(headers)], " | "))
println(join([repeat("-", w) for w in widths], "-+-"))
for row in summary_rows
  println(join([rpad(string(getters[j](row)), widths[j]) for j in_
    ↪eachindex(getters)], " | "))
end

println("\nProtein Vectors (%; ivec in columns)")
protein_categories = [c for c in protein_order if haskey(protein_by_cat, c)]
print_pivot(protein_categories, protein_by_cat, ivec_list)

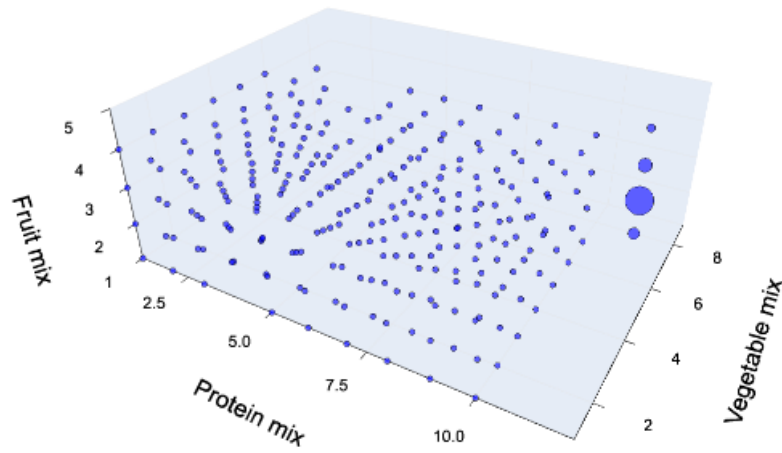
println("\nVegetable Vectors (%; ivec in columns)")
veg_categories = [c for c in veg_order if haskey(veg_by_cat, c)]
print_pivot(veg_categories, veg_by_cat, ivec_list)

```

--- Adjoint stratification with ivec = 2 ---

[Info: Found 4 derivations for stratification.

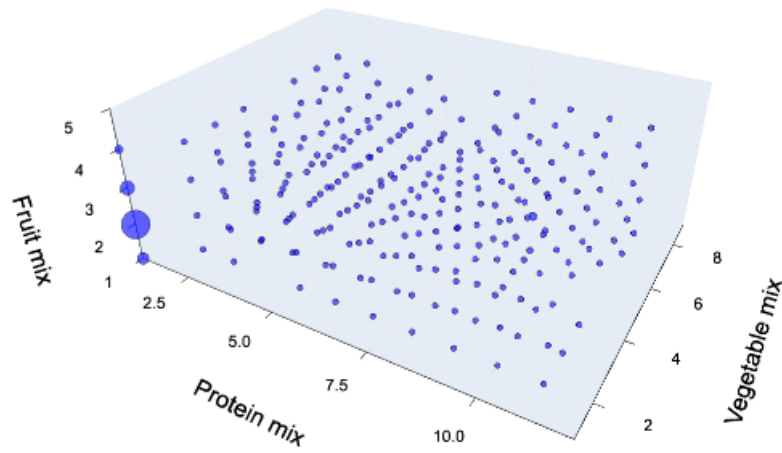
Adjoint Stratified Tensor (ivec=2)



--- Adjoint stratification with ivec = 3 ---

[Info: Found 4 derivations for stratification.

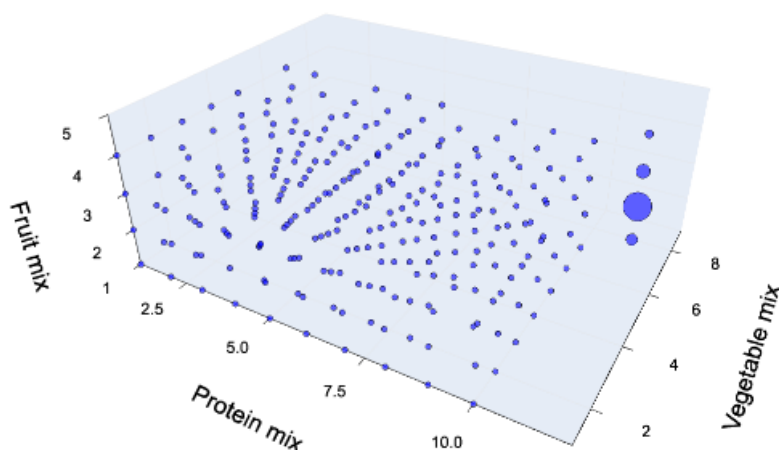
Adjoint Stratified Tensor (ivec=3)



--- Adjoint stratification with ivec = 4 ---

[Info: Found 4 derivations for stratification.

Adjoint Stratified Tensor (ivec=4)



Fixed Cluster Table (Adjoint ivec=2,3,4)

ivec	requested	used
2	(p=>11, v=>8)	(p=>11, v=>8)
3	(p=>1, v=>1)	(p=>1, v=>1)
4	(p=>11, v=>8)	(p=>11, v=>8)

Protein Vectors (%; ivec in columns)

category	ivec=2	ivec=3	ivec=4
Meat (beef, pork, lamb, game)	30.1%	25.5%	2.4%
Cured meat	19.7%	-9.2%	-4.2%
Organ meats	-3.6%	-3.6%	96.0%
Poultry	34.1%	20.3%	1.2%
High omega-3 seafood	0.4%	-2.5%	23.2%
Low omega-3 seafood	14.3%	-3.1%	-1.6%
Eggs	48.2%	66.1%	5.0%
Soy products	-2.4%	0.8%	13.8%
Nuts and seeds	60.4%	-59.9%	0.6%
Legumes (as protein)	-22.8%	29.5%	-1.2%
No measurable protein	-28.9%	0.6%	-1.6%

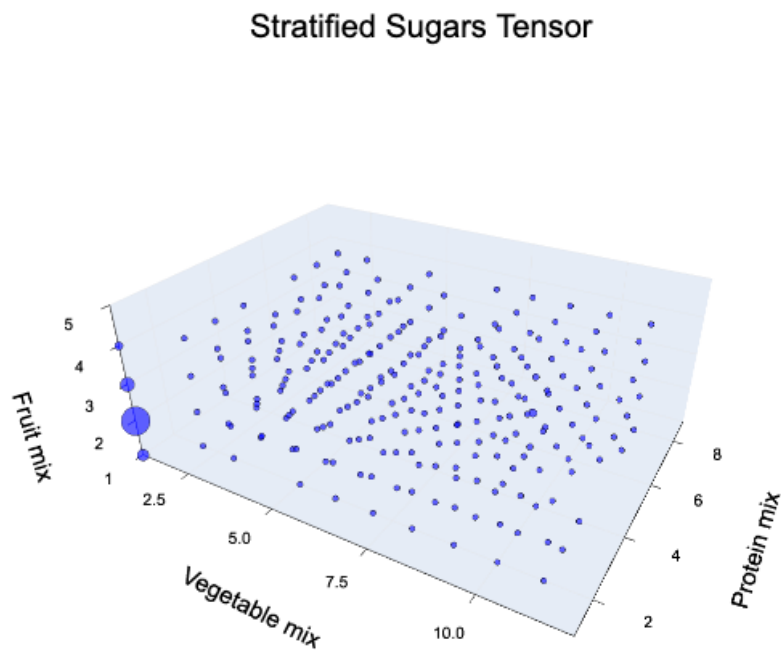
Vegetable Vectors (%; ivec in columns)

category	ivec=2	ivec=3	ivec=4
Dark green vegetables	74.5%	74.5%	74.5%
Red/orange vegetables (tomatoes)	0.3%	0.3%	0.3%
Red/orange vegetables (other)	-1.7%	-1.7%	-1.7%
Starchy vegetables (potatoes)	-53.7%	-53.7%	-53.7%
Starchy vegetables (other)	39.5%	39.5%	39.5%
Other vegetables	-1.1%	-1.1%	-1.1%
Legumes (as vegetables)	0.4%	0.4%	0.4%

Adjoints have 1 trivial solution so even in the adjoint position there 3 feature vectors (less than the 5 for the universal chisel).

```
[153]: inds_adj = collect(ITensors.inds(Sugar_adj_strat))
plot_tensor(Sugar_adj_strat, title="Stratified Sugars Tensor",
            xlabel=mode_label(inds_adj[2]; mixed=true),
            ylabel=mode_label(inds_adj[1]; mixed=true), zlabel=mode_label(inds_adj[3];
            mixed=true))
```

[153]:



3.1 Summary

While this experiment is for demonstration purposes and we are not including subject matter expertise we can draw the following conclusions from the computation.

1. Tucker decompositions at the default tolerance are trivial.
2. Higher Order SVDs do not appear to locate a specific cluster in fact they make the data geometrically more dense than in the given representation.
3. The Dleto universal chisel identifies 5 independent derivations. Their applied null patterns identify a block in the protein-vegetable modes. Further clustering structure is visible but less detached but hints at more nuanced clustering.
4. The clustering is visible even in the adjoint chisel but with only 3 independent derivations so it features more structure but not as much as with universal chiselling.
5. Analyzing the main cluster, the fruit mode is ignored, the mixtures supports combinations of
 - proteins that are strong in meat, cured meat, and poultry;
 - vegetables that are a strong mix of dark greens and starchy vegetables that are not potatoes.

Without subject matter knowledge we can only guess what is the source of the added sugars in these foods. Perhaps these are in the form of salad dressings or sauces? We can only conjecture. The point is however that the data here encodes a signal to pursue that is evident with the chiselling method and at least not transparently noticed by other rank related strategies. Since the outlier here is small dimension (1-dimensional) it is understandable that its influence on singles such as singular values will be muted.